SIMATS ENGINEERING
ASSESSMENT TOOL 2

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COURSE NAME: CSA1603 **COURSE CODE:** Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario ***(2)***	Identification of Key Issues ***(2)***	Application of Concepts ***(2.5)***	Decision Making ***(2)***	Clarity of Response ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

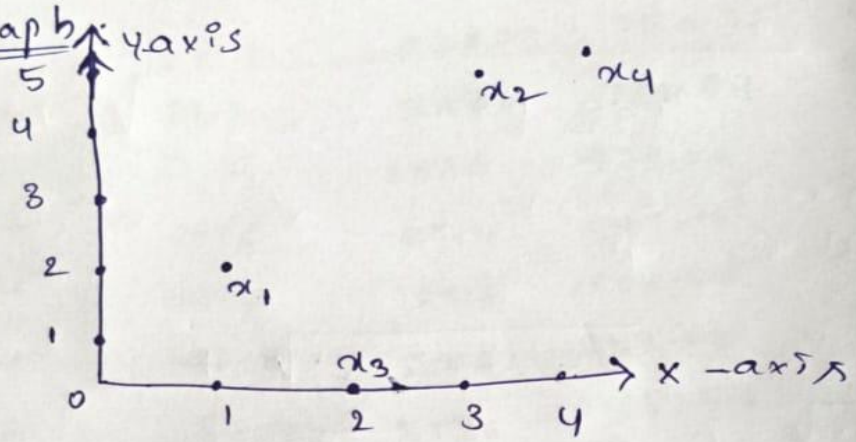
I _ M A Surya Kiran (192411236)*_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

M A Surya Kiran
Signature of the student:

Minkowski Distance Activity (5)

① * Consider the following points : (1,2), (3,5), (2,0) & (4,5) construct th data matrix & similarity matrix Manhatta Euclidean Distance.

Graph :



Data Matrix (output)

Point	attribute 1	attribute 2
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

Dissimilarity Matrix Using Manhattan :

	x_1	x_2	x_3	x_4
x_1	0	×	×	×
x_2	5	0	×	×
x_3	3	6	0	×
x_4	6	1	7	0

(i) x_2, x_1 ; $x_2 = (3, 5)$ $x_1 = (1, 2)$

$x_1\ y_1$ $x_2\ y_2$

$$d = (|x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}|)$$

$$d = 2 + 3 \Rightarrow \boxed{5}$$

(ii) x_3, x_1 ; $x_3 = (2, 0)$, $x_1 = (1, 2)$

$$d = (|x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}|)$$

$$d = ((1) + (|-2|)$$

$$d = (1 + 2) \Rightarrow \boxed{3}$$

(iii) x_4, x_1 ; $x_4 = (4, 5)$, $x_1 = (1, 2)$

$$d = (|x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}|)$$

$$d = (|3|) + (|3|)$$

$$\boxed{d \Rightarrow 6}$$

(iv) x_3, x_2 ; $x_3 = (2, 0)$; $x_2 = (3, 5)$

$$d = (1) + (5)$$

$$\boxed{d = 6}$$

(v) x_4, x_3 ; $x_4 = (4, 5)$, $x_3 = (2, 0)$

$$d = (2) + (5)$$

$$\boxed{d = 7}$$

(vi) x_4, x_2 ; $x_4 = (4, 5)$, $x_2 = (3, 5)$

$$d = (1) + (0)$$

$$\boxed{d = 1}$$

Dissimilarity matrix by using Euclidean

	x_1	x_2	x_3	x_4
x_1	0	×	×	×
x_2	3.61	0	×	×
x_3	2.24	5.1	0	×
x_4	4.24	1	5.39	0

i) $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

$x_2, x_1 \Rightarrow x_2 = (3,5)$ $x_1 = (1,2)$
$x_1\ y_1 \quad x_2, y_2$

$d = \sqrt{(3-1)^2 + (5-2)^2}$

$d = \sqrt{(2)^2 + (3)^2} \Rightarrow d = \sqrt{4+9} \Rightarrow \sqrt{13}$

$\boxed{d \Rightarrow 3.605}$

(ii) x_3, x_1 $x_3 = (2,0)$, $x_1 = (1,2)$

$d = \sqrt{(2-1)^2 + (0-2)^2} \Rightarrow \sqrt{1+4} \Rightarrow \sqrt{5}$

$\boxed{d = 2.24}$

(iii) x_3, x_2 $x_3 = (2,0)$, $x_2 = (3,5)$

$d = \sqrt{(-1)^2 + (-5)^2} \Rightarrow \sqrt{1+25} \Rightarrow \sqrt{26}$

$\boxed{d = 5.1}$

(iv) x_4, x_1 $x_4 = (4,5)$, $x_1 = (1,2)$

$d = \sqrt{(3)^2 + (3)^2} \Rightarrow \sqrt{18} \Rightarrow \boxed{4.24} = d$

(v) $x_4, x_2 \Rightarrow \sqrt{(1)^2 + (0)^2} \Rightarrow \sqrt{1} \Rightarrow \boxed{1}$

(vi) $x_4, x_3 \Rightarrow \sqrt{(2)^2 + (5)^2} \Rightarrow \sqrt{29} \Rightarrow \boxed{5.39}$

Dissimilarity matrix by Using Supremum

L_∞	x_1	x_2	x_3	x_4
x_1	0			
x_2	3	0		
x_3	2	5	0	
x_4	3	1	5	0

$d(x_i, x_j) \Rightarrow \max\,(x_1 - x_2),\ (y_1 - y_2)$

(i) $x_2 = (3,5)$, $x_1 = (1,2)$

$|3-1|, |5-2|$

$d(x_2, x_1) = \max(2,3) \Rightarrow \boxed{3}$

(ii) (x_3, x_1) $x_3 = (2,0)$, $x_1 = (1,2)$

$|2-1|, |0-2|$

$d(x_3, x_1) = \max(1,2) \Rightarrow \boxed{2}$

(iii) (x_3, x_2) $x_3 = (2,0)$ $x_2 = (3,5)$

$|2-3|, |0-5|$

$d(x_3, x_2) = \max(1,5) \rightarrow \boxed{5}$

(iv) $(x_4, x_1) \Rightarrow x_4 = (4,5)$ $x_1 (1,2)$

$|4-1|, |5-2|$

$d(x_4, x_1) = \max(3,3) \Rightarrow \boxed{3}$

(v) $(x_4, x_2) \Rightarrow x_4 = (4,5)$ $x_2 = (3,5)$

$|4-3|, |5-5|$

$d(x_4, x_2) = \max(1,0) \Rightarrow \boxed{1}$

(vi) $(x_4, x_3) \Rightarrow x_4 = (4,5)$, $x_3 = (2,0)$

$|4-2|, |5-0|$

$d(x_4, x_3) = \max(2,5) \Rightarrow \boxed{5}$

② Finding the Similarity measure between two documents Using cosine Similarity

Document	Team	coach	hockey	baseball	Soccer	penalty
Document 1	5	0	3	0	2	0
Document 2	3	0	2	0	1	1
Document 3	0	7	0	2	1	0
Document 4	0	1	0	0	1	2

Document	Score	win	loss	Season
Document 1	0	2	0	0
Document 2	0	1	0	1
Document 3	0	3	0	0
Document 4	2	0	3	0

Formula:

$$\cos(d_1, d_2) = \frac{(d_1 \cdot d_2)}{||d_1||\ ||d_2||}$$

$\therefore ||d|| \rightarrow$ Length of Vector

$\therefore \cdot \rightarrow$ Vector dot product

Finding similarity btw document d_1 & d_2.

$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$

$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$

$d_1 \cdot d_2 \Rightarrow ((5 \cdot 3), 0, (3 \cdot 2), 0, (2 \cdot 1), (0 \cdot 1), (0), (2 \cdot 1), (0), (0 \cdot 1))$

$\Rightarrow ((15), 0, (6), 0, (2), 0, 0, (2), 0, 0)$

$\Rightarrow (15, 6, 2, 2)$

$d_1 \cdot d_2 \Rightarrow 25$

$$|d_1| \Rightarrow 25+9+4+4 \quad (5^2, 3^2, 2^2, 2^2)$$

$$\Rightarrow 42$$

$$||d_1|| \Rightarrow \sqrt{42} \Rightarrow \boxed{6.481 = ||d_1||}$$

$$|d_2| \Rightarrow (3^2, 2^2, 1^2, 1^2, 1^2, 1^2)$$

$$\Rightarrow (9+4+1+1+1+1)$$

$$\Rightarrow 17$$

$$||d_2|| \Rightarrow \sqrt{17} \Rightarrow \boxed{4.12 = ||d_2||}$$

$$\cos(d_1, d_2) = \frac{(25)}{(6.481)(4.12)}$$

$$\Rightarrow \frac{25}{26.70172} \Rightarrow 0.934$$

$$\boxed{\cos(d_1 d_2) = 0.934} \approx 0.94$$

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Scenario based Questions

Q1. Data Preprocessing in Retail Analytics *(L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	**Age**	**Gender**	**Monthly_Spend (₹K)**	**Purchase_Frequency**	**Loyalty_Level**
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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B. Outlier Detection and Treatment

i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization

i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High

- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

📊 Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

✅ Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records
ii. Assign each value to its respective bin
iii. Analyze how effectively this method groups short and long visit durations

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B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration
ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

ii. Transform all values accordingly
iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**